

word embedding $E_\ell[i] \in \mathbb{R}^d$ as a linear combination of English word embeddings $E_e[j] \in \mathbb{R}^d$

$$E_\ell[i] = \sum_{j=1}^{|V_e|} \alpha_{ij} E_e[j] = \alpha_i E_e \quad (1)$$

where $\alpha_i \in \mathbb{R}^{|V_e|}$ is a sparse vector and $\sum_j \alpha_{ij} = 1$.

In this step of initializing foreign embeddings, having a good estimation of α could speed of the convergence when tuning the foreign model and enable zero-shot transfer (§5). In the following, we discuss how to estimate $\alpha_i \forall i \in \{1, 2, \dots, |V_\ell|\}$ under two scenarios: (i) we have parallel data of English-foreign, and (ii) we only rely on English and foreign monolingual data.

Learning from Parallel Corpus Given an English-foreign parallel corpus, we can estimate word translation probability $p(e_j | \ell_i)$ for any (English-foreign) pair (e_j, ℓ_i) using popular word-alignment (Brown et al., 1993) toolkits such as fast-align (Dyer et al., 2013). We then assign

$$\alpha_{ij} = p(e_j | \ell_i) \quad (2)$$

Since α_i is estimated from word alignment, it is a sparse vector.

Learning from Monolingual Corpus For low resource languages, parallel data may not be available. In this case, we rely only on monolingual data (e.g., Wikipedias). We estimate word translation probabilities from word embeddings of the two languages. Word vectors of these languages can be learned using fastText (Bojanowski et al., 2017) and then are aligned into a shared space with English (Lample et al., 2018b; Joulin et al., 2018). Unlike learning contextualized representations, learning word vectors is fast and computationally cheap. Given the aligned vectors $\bar{E}_\ell$ of foreign and $\bar{E}_e$ of English, we calculate the word translation matrix $A \in \mathbb{R}^{|V_\ell| \times |V_e|}$ as

$$A = \text{sparsemax}(\bar{E}_\ell \bar{E}_e^\top) \quad (3)$$

Here, we use sparsemax (Martins and Astudillo, 2016) instead of softmax. sparsemax is a sparse version of softmax and it puts zero probabilities on most of the words in the English vocabulary except few English words that are similar to a given foreign word. This property is desirable in our approach since it leads to a better initialization of the foreign word embeddings.

2.2 Fine-tuning Bilingual LM

We create a bilingual LM by plugging foreign language specific parameters to the pretrained English LM (Figure 1). The new model has two separate word embedding layers (and output layers), one for English and one for foreign language. The encoder layer in between is shared. We then fine-tune this model using English and foreign monolingual data. Here, we keep tuning the model on English to ensure that it does not forget what it has learned in English and that we can use the resulting model for zero-shot transfer (§3). In this step, the encoder is updated so that it can learn syntactic aspects (i.e., word order, morphological agreement) of the target languages.

3 Zero-shot Experiments

In the scope of this work, we focus on transferring autoencoding LMs trained with masked language model objective. We choose BERT and RoBERTa (Liu et al., 2019) as the source models for building our bilingual language models, named RAMEN³ for the ease of discussion. We implement our models on top of HuggingFace’s Transformer (Wolf et al., 2019).⁴ For each pretrained model, we experiment with 12 layers (BERT_{BASE} and RoBERTa_{BASE}) and 24 layers (BERT_{LARGE} and RoBERTa_{LARGE}) variants. Using BERT_{BASE} allows us to compare the results with mBERT model. Using BERT_{LARGE} and RoBERTa allows us to investigate whether the performance of the target LM correlates with the performance of the source pretrained model. RoBERTa is a recently published model that is similar to BERT architecturally but with an improved training procedure. By training for longer time, with bigger batches, on more data, and on longer sequences, RoBERTa matched or exceeded previously published models including XLNet. We include RoBERTa in our experiments to validate the motivation of our work: *with similar architecture, does a stronger pretrained English model result in a stronger bilingual LM?* We evaluate our models on two cross-lingual zero-shot tasks: (1) Cross-lingual Natural Language Inference (XNLI) and (2) dependency parsing.

³The author likes ramen.

⁴<https://github.com/huggingface/transformers>